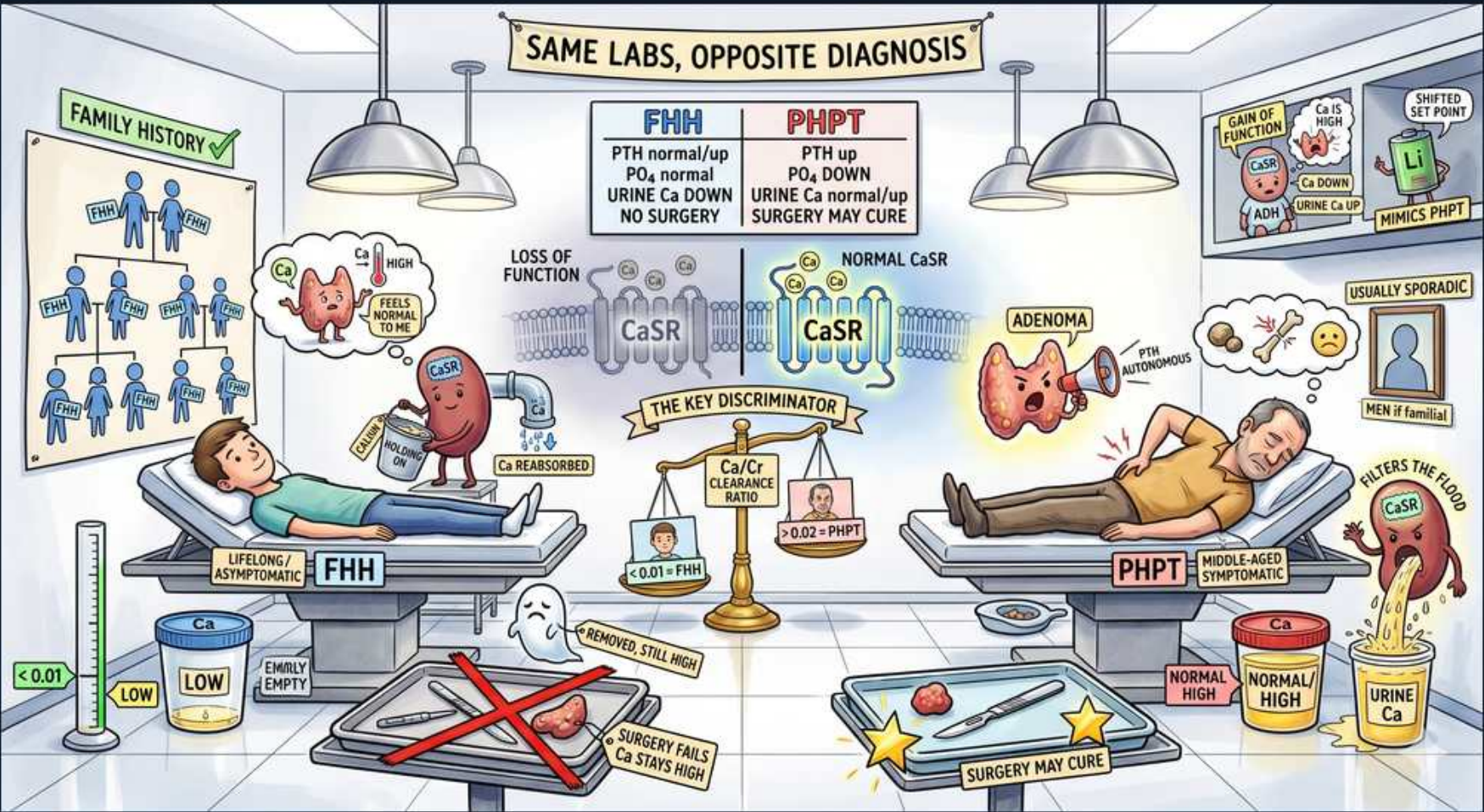


Calcium/PTH/VitD — The CaSR Sensitivity Lab (FHH vs PHPT)



1. Family history checkmark

WHAT YOU SEE

A green 'FAMILY HISTORY' banner with a chart of multiple affected family members.

WHAT IT ENCODES

FHH is autosomal dominant, so multiple relatives across generations have lifelong, asymptomatic, mild hypercalcemia. A positive family history of benign hypercalcemia (and especially of a FAILED prior parathyroidectomy in a relative) strongly favors FHH over sporadic primary hyperparathyroidism. Onset is present from birth/childhood rather than discovered in middle age.

BOARD TRAP

WRONG to assume any hereditary hypercalcemia with high-normal PTH must be MEN-related primary HPT and order RET/MEN1 testing first — FHH is benign, far more common in this exact lab picture, and needs no surgery. Compute the urine Ca/Cr clearance ratio before pursuing a syndromic workup.

2. FHH label panel

WHAT YOU SEE

Left blue header box titled FHH listing PTH normal/up, PO4 normal, urine Ca DOWN, NO SURGERY — the column header that fingerprints the benign hereditary disorder (note the low urine calcium and normal phosphate).

WHAT IT ENCODES

FHH (familial hypocalciuric hypercalcemia): mild hypercalcemia present since birth, inappropriately normal or mildly elevated PTH, NORMAL phosphate, and characteristically LOW urine calcium. Magnesium is often high-normal. Because the receptor defect is body-wide and the condition is benign, the management is reassurance and NO surgery, NO drug therapy.

BOARD TRAP

WRONG to expect LOW phosphate in FHH the way you would in primary HPT — in FHH phosphate is typically normal, and urine calcium is LOW (the kidney's loss-of-function CaSR avidly reabsorbs calcium). Low urine calcium + normal phosphate is the FHH fingerprint; low phosphate + higher urine calcium is primary HPT.

3. PHPT label panel

WHAT YOU SEE

Right red header box titled PHPT listing PTH up, PO4 DOWN, urine Ca normal/up, SURGERY MAY CURE — the contrasting column header where low phosphate and a curable adenoma set it apart from FHH.

WHAT IT ENCODES

Primary hyperparathyroidism: HIGH calcium, frankly HIGH (or inappropriately normal) PTH, LOW phosphate (PTH-driven renal wasting), and NORMAL-to-HIGH urine calcium. Most often a single adenoma in a middle-aged/postmenopausal woman found incidentally. Parathyroidectomy is curative when indicated.

BOARD TRAP

WRONG to conclude there is 'no parathyroid problem' when PTH sits in the normal range alongside high calcium — a non-suppressed PTH during hypercalcemia is inappropriate and defines primary HPT. Read PTH relative to the calcium level, not against the reference interval.

4. Loss-of-function CaSR (left)

WHAT YOU SEE

A receptor labeled CaSR with 'LOSS OF FUNCTION' over the FHH side.

WHAT IT ENCODES

FHH is caused by heterozygous INACTIVATING (loss-of-function) mutations in CASR (the G-protein-coupled calcium-sensing receptor). The receptor under-senses serum calcium, so it takes a higher-than-normal calcium level to suppress PTH — the calcium set-point is shifted upward. The same defective receptor in the renal tubule increases calcium reabsorption, producing the low urine calcium.

BOARD TRAP

WRONG to confuse this with ACTIVATING (gain-of-function) CaSR mutations, which cause autosomal dominant HYPOcalcemia (the receptor over-senses calcium, suppresses PTH too readily, and the kidney wastes calcium) — the exact opposite disease, and one where calcium supplementation/calcitriol can worsen hypercalciuria. Direction of the mutation flips the whole phenotype.

5. Normal CaSR (right)

WHAT YOU SEE

A second CaSR receptor labeled 'NORMAL CaSR' on the PHPT side.

WHAT IT ENCODES

In primary HPT the calcium-sensing receptor itself is intrinsically NORMAL; the problem is autonomous PTH secretion from an adenoma or hyperplasia that ignores normal feedback. Because the renal CaSR works, the kidney does NOT avidly reclaim calcium, so urine calcium is normal-to-high — the basis for the discriminating clearance ratio.

BOARD TRAP

WRONG to think a calcimimetic (cinacalcet) is futile in primary HPT because 'the receptor is normal' — in fact a normal CaSR is exactly what cinacalcet sensitizes to lower PTH/Ca when surgery is declined. In FHH (defective receptor) cinacalcet is less reliably effective; the receptor status predicts drug response.

6. Parathyroid 'feels normal' (FHH)

WHAT YOU SEE

A parathyroid gland cartoon saying it feels Ca is normal and reabsorbs Ca.

WHAT IT ENCODES

With a loss-of-function CaSR in BOTH the parathyroid and the kidney, the body misreads true hypercalcemia as 'calcium is normal/low.' The parathyroid therefore keeps PTH inappropriately normal/high, and the renal tubule keeps reabsorbing calcium (hypocalciuria). This dual-organ misperception explains why FHH has high Ca, non-suppressed PTH, AND low urine Ca simultaneously.

BOARD TRAP

WRONG to expect that fixing or removing the parathyroid glands corrects FHH — the RENAL CaSR defect persists and continues reabsorbing calcium, so calcium stays high after parathyroidectomy. The body-wide nature of the receptor defect is why surgery fails.

7. Parathyroid adenoma (PHPT)

WHAT YOU SEE

An enlarged red parathyroid adenoma labeled 'ADENOMA / PTH autonomous'.

WHAT IT ENCODES

A single autonomous adenoma causes ~80–85% of primary HPT and secretes PTH independent of calcium feedback. Localize before minimally invasive surgery with sestamibi SPECT/CT and/or neck ultrasound; intraoperative PTH dropping >50% within 10 minutes confirms removal of the culprit gland.

BOARD TRAP

WRONG to default to single-adenoma surgery in a young patient with multigland disease — four-gland HYPERPLASIA points to a familial/MEN syndrome (MEN1: parathyroid + pituitary + pancreas; MEN2A: parathyroid + medullary thyroid cancer + pheochromocytoma) and requires subtotal (3.5-gland) parathyroidectomy. Single vs four-gland disease changes the operation.

8. MEN if familial badge

WHAT YOU SEE

A small cartoon with 'MEN if familial' and 'usually sporadic'.

WHAT IT ENCODES

Primary HPT is usually sporadic (single adenoma). Familial/multigland disease or young age should raise suspicion for MEN1 (MEN1 menin gene) or MEN2A (RET proto-oncogene). In MEN2A, pheochromocytoma must be excluded and resected BEFORE any neck or parathyroid surgery to avoid intraoperative hypertensive crisis.

BOARD TRAP

WRONG to proceed to parathyroid/thyroid surgery in a possible MEN2A patient without first screening for and treating pheochromocytoma — operating with an unrecognized pheo can precipitate a fatal catecholamine crisis. Always rule out pheo before neck surgery in suspected MEN2A.

9. Ca/Cr clearance ratio scale

WHAT YOU SEE

A balance labeled 'THE KEY DISCRIMINATOR — Ca/Cr CLEARANCE RATIO'.

WHAT IT ENCODES

The 24-hour calcium-to-creatinine CLEARANCE ratio is the single best discriminator: $\text{ratio} = (\text{urine Ca} \times \text{serum Cr}) / (\text{serum Ca} \times \text{urine Cr})$. It normalizes urinary calcium to renal function. A value <0.01 favors FHH (hypocalciuric); >0.02 favors primary HPT. Always confirm vitamin D is replete first, because deficiency can falsely lower urine calcium.

BOARD TRAP

WRONG to use a raw 24-h urine calcium or spot urine Ca instead of the calculated CLEARANCE ratio — absolute urine calcium varies with intake and GFR and can mislead. The ratio formula corrects for both. Using the wrong metric is how FHH patients get mistakenly sent to surgery.

10. <0.01 = FHH

WHAT YOU SEE

A green tag reading '< 0.01 = FHH' hanging off the low pan of the balance — the cut-off marking hypocalciuria, because the broken renal CaSR hoards calcium and barely spills any into urine.

WHAT IT ENCODES

A calcium-to-creatinine clearance ratio <0.01 strongly supports FHH — the loss-of-function CaSR in the kidney avidly reabsorbs calcium, so very little is excreted (hypocalciuria) despite high serum calcium. Genetic CASR testing can confirm when the ratio is intermediate or the diagnosis is unclear.

BOARD TRAP

WRONG to interpret a ratio in the indeterminate 0.01–0.02 gray zone as definitive for either disease — about 20% of FHH and some primary HPT fall here, and concurrent vitamin D deficiency or thiazide use can lower the ratio. In the gray zone, recheck after vitamin D repletion and consider CASR genetic testing.

11. >0.02 = PHPT

WHAT YOU SEE

A tag reading '> 0.02 = PHPT' on the other pan of the balance — the cut-off marking normal/high urine calcium, because the intact renal CaSR lets the high filtered calcium spill out (and form stones).

WHAT IT ENCODES

A clearance ratio >0.02 supports primary HPT: the normal renal CaSR does not avidly reclaim calcium, so the high filtered calcium load spills into the urine (normal/high urine calcium). This hypercalciuria is also why primary HPT patients form calcium kidney stones — a 'stones' feature absent in FHH.

BOARD TRAP

WRONG to assume that high urine calcium alone proves primary HPT — thiazides (lower urine Ca) or coexisting conditions can shift it, and the ratio must still be calculated and interpreted with serum Ca/PTH. The discriminator is the calculated ratio plus the whole biochemical picture, not urine calcium in isolation.

12. FHH patient — lifelong asymptomatic

WHAT YOU SEE

A young patient lying down labeled 'FHH — LIFELONG ASYMPTOMATIC'.

WHAT IT ENCODES

FHH patients are typically young, lifelong asymptomatic, and have NO end-organ damage from calcium — no stones, no osteoporosis, no neuropsychiatric crisis. Because it is benign, the correct management is reassurance and avoiding unnecessary surgery; severe neonatal hyperparathyroidism (homozygous CASR) is the rare exception requiring urgent treatment.

BOARD TRAP

WRONG to treat or operate on an asymptomatic FHH patient to 'normalize' the calcium — intervention adds risk without benefit because the hypercalcemia is harmless and surgery does not cure it. Recognizing FHH spares the patient an unnecessary operation.

13. PHPT patient — middle-aged symptomatic

WHAT YOU SEE

An older patient clutching flank, labeled 'PHPT — MIDDLE-AGED SYMPTOMATIC'.

WHAT IT ENCODES

Primary HPT presents in middle-aged/postmenopausal patients and can cause end-organ damage: nephrolithiasis/nephrocalcinosis (the flank pain shown), osteoporosis (especially cortical bone at the distal radius), abdominal symptoms, and neuropsychiatric complaints — 'stones, bones, groans, and psychiatric moans.' These complications drive the indications for curative parathyroidectomy.

BOARD TRAP

WRONG to attribute a middle-aged stone-former's hypercalcemia to FHH just because PTH is high-normal — FHH does NOT cause stones or osteoporosis. Symptomatic end-organ disease plus a clearance ratio >0.02 points firmly to primary HPT and surgical evaluation.

14. Surgery fails / red X in FHH

WHAT YOU SEE

A red X over a removed gland: 'REMOVED, STILL HIGH — SURGERY FAILS, Ca STAYS HIGH'.

WHAT IT ENCODES

Parathyroidectomy does NOT cure FHH because the calcium-sensing defect is body-wide (parathyroid AND renal tubule), not confined to the gland — after removing parathyroid tissue, the renal CaSR keeps reabsorbing calcium and serum calcium stays elevated. This is why diagnosing FHH BEFORE any referral is critical.

BOARD TRAP

WRONG to escalate to repeat neck exploration when calcium stays high after a 'parathyroidectomy' — persistent hypercalcemia with low urine calcium should make you reconsider FHH (a missed diagnosis), not assume residual adenoma. Operating on FHH is the signature board trap; the urine Ca/Cr clearance ratio prevents it.